

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
cat('sdev = 0.005, beta.var.scale = 0.2\n')
```

```
## sdev = 0.005, beta.var.scale = 0.2
```

Key Summary Statistics

```
cat(paste0('```\n', '
--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.6067  0.8283  0.8433  0.8311  0.8522  0.8728

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.08833 0.13389 0.12909 0.16667 0.86611
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.09056 0.13667 0.13173 0.17000 0.72333
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.005556 0.009444 0.104215 0.047778 1.000000
```

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Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0005556 0.0055556 0.0083333 0.0776651 0.0262500 1.0000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0083333 0.036667 0.182825 0.202222 1.000000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.4867  0.5678  0.6183  0.6222  0.6689  0.8111

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.04444 0.08667 0.09349 0.12778 0.85944
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.04667 0.09000 0.09549 0.13111 0.66667
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.001111 0.056667 0.315982 0.682222 1.000000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.4678  0.7050  0.7589  0.7631  0.8194  0.9922
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0005556 0.0033333 0.1687351 0.1626389 1.0000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.0000  0.1722  0.6411  0.5646  0.9394  1.0000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.9628  0.9928  0.9950  0.9943  0.9967  1.0000

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02611 0.05889 0.06933 0.09611 0.90056
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02778 0.06111 0.07151 0.09944 0.60444
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0001832 0.0000000 0.8344444
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000e+00 0.000e+00 0.000e+00 5.054e-05 0.000e+00 4.889e-02
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0001509 0.0000000 0.0527778
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0002125 0.0000000 0.5483333

--- Summary for K = 4 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.9750 0.9928 0.9944 0.9943 0.9961 1.0000

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01611 0.04000 0.05343 0.07333 0.87611
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

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    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1        1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01681 0.04222 0.05522 0.07667 0.64056
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000000 0.0000000 0.0000000 0.0000289 0.0000000 0.6377778
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000e+00 0.000e+00 0.000e+00 2.572e-06 0.000e+00 8.889e-03
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000e+00 0.000e+00 0.000e+00 3.897e-06 0.000e+00 3.889e-03
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000e+00 0.000e+00 0.000e+00 4.035e-06 0.000e+00 8.333e-03

--- Summary for K = 5 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.9244  0.9917  0.9939  0.9932  0.9956  0.9989

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01222 0.03167 0.04878 0.06333 0.88111
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1        1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1        1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1        1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1        1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01333 0.03333 0.04955 0.06667 0.52278
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000e+00 0.000e+00 0.000e+00 2.920e-06 0.000e+00 6.556e-02
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000e+00 0.000e+00 0.000e+00 6.660e-07 0.000e+00 3.889e-03
  Modality 4 :

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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.897e-06	0.000e+00	6.111e-03

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0000399	0.0000000	0.8722222

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9378	0.9911	0.9933	0.9926	0.9950	0.9989

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.008889	0.023333	0.041989	0.050000	0.897778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.01000	0.02444	0.04190	0.05333	0.52778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0000088	0.0000000	0.3366667

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	7.560e-08	0.000e+00	1.111e-03

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8889	0.9589	0.9728	0.9694	0.9850	0.9972

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
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0.000000 0.006667 0.016667 0.030745 0.037778 0.852778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.017778	0.031441	0.040000	0.456667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0000277	0.0000000	0.9922222

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.002214	0.000000	1.000000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00e+00	0.00e+00	0.00e+00	4.86e-05	0.00e+00	1.00e+00

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9806	0.9917	0.9933	0.9932	0.9950	0.9989

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.013889	0.023370	0.029444	0.698889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

```
gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.005556 0.014444 0.024619 0.031111 0.695556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    0         0         0         0         0         0
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    0         0         0         0         0         0
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    0         0         0         0         0         0
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.00e+00 0.00e+00 0.00e+00 4.28e-05 0.00e+00 1.00e+00
```

--- Summary for K = 9 ---

```
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.9828 0.9917 0.9933 0.9932 0.9950 0.9994
```

```
beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.004444 0.011111 0.018720 0.023333 0.836667
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
```

```
gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.005556 0.012222 0.019669 0.025556 0.542222
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000e+00 0.000e+00 0.000e+00 2.430e-06 0.000e+00 1.111e-01
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    0         0         0         0         0         0
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    0         0         0         0         0         0
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
```

```

0 0 0 0 0 0
', '\n'))

```

```

## ' '
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6067  0.8283  0.8433  0.8311  0.8522  0.8728
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.08833 0.13389 0.12909 0.16667 0.86611
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.09056 0.13667 0.13173 0.17000 0.72333
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005556 0.009444 0.104215 0.047778 1.000000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0005556 0.0055556 0.0083333 0.0776651 0.0262500 1.0000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.008333 0.036667 0.182825 0.202222 1.000000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4867  0.5678  0.6183  0.6222  0.6689  0.8111
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```



```

## 0.00000 0.04444 0.08667 0.09349 0.12778 0.85944
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.04667 0.09000 0.09549 0.13111 0.66667
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.001111 0.056667 0.315982 0.682222 1.000000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.4678 0.7050 0.7589 0.7631 0.8194 0.9922
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000000 0.0005556 0.0033333 0.1687351 0.1626389 1.0000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.1722 0.6411 0.5646 0.9394 1.0000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.9628 0.9928 0.9950 0.9943 0.9967 1.0000
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.02611 0.05889 0.06933 0.09611 0.90056
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02778 0.06111 0.07151 0.09944 0.60444
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0001832 0.0000000 0.8344444
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 5.054e-05 0.000e+00 4.889e-02
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0001509 0.0000000 0.0527778
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0002125 0.0000000 0.5483333
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9750 0.9928 0.9944 0.9943 0.9961 1.0000
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01611 0.04000 0.05343 0.07333 0.87611
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01681 0.04222 0.05522 0.07667 0.64056
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0000289 0.0000000 0.6377778
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 2.572e-06 0.000e+00 8.889e-03
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 3.897e-06 0.000e+00 3.889e-03
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 4.035e-06 0.000e+00 8.333e-03

```

```

##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9244 0.9917 0.9939 0.9932 0.9956 0.9989
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01222 0.03167 0.04878 0.06333 0.88111
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01333 0.03333 0.04955 0.06667 0.52278
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 2.920e-06 0.000e+00 6.556e-02
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 6.660e-07 0.000e+00 3.889e-03
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 3.897e-06 0.000e+00 6.111e-03
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0000399 0.0000000 0.8722222
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9378 0.9911 0.9933 0.9926 0.9950 0.9989
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.008889 0.023333 0.041989 0.050000 0.897778
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```

##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01000 0.02444 0.04190 0.05333 0.52778
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0000088 0.0000000 0.3366667
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 7.560e-08 0.000e+00 1.111e-03
##
## --- Summary for K = 7 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.8889 0.9589 0.9728 0.9694 0.9850 0.9972
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.006667 0.016667 0.030745 0.037778 0.852778
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.007778 0.017778 0.031441 0.040000 0.456667
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0000277 0.0000000 0.9922222

```

```

## Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
## 0.000000 0.000000 0.000000 0.002214 0.000000 1.000000
## Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
##      0      0      0      0      0      0
## Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
## 0.00e+00 0.00e+00 0.00e+00 4.86e-05 0.00e+00 1.00e+00
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
##  0.9806  0.9917  0.9933  0.9932  0.9950  0.9989
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
## 0.000000 0.005556 0.013889 0.023370 0.029444 0.698889
## Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
## 0.000000 0.005556 0.014444 0.024619 0.031111 0.695556
## Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
##      0      0      0      0      0      0
## Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
##      0      0      0      0      0      0
## Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
##      0      0      0      0      0      0
## Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
## 0.00e+00 0.00e+00 0.00e+00 4.28e-05 0.00e+00 1.00e+00
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
##  0.9828  0.9917  0.9933  0.9932  0.9950  0.9994
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.004444 0.011111 0.018720 0.023333 0.836667
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.005556 0.012222 0.019669 0.025556 0.542222
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 2.430e-06 0.000e+00 1.111e-01
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##
## ‘‘‘

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.